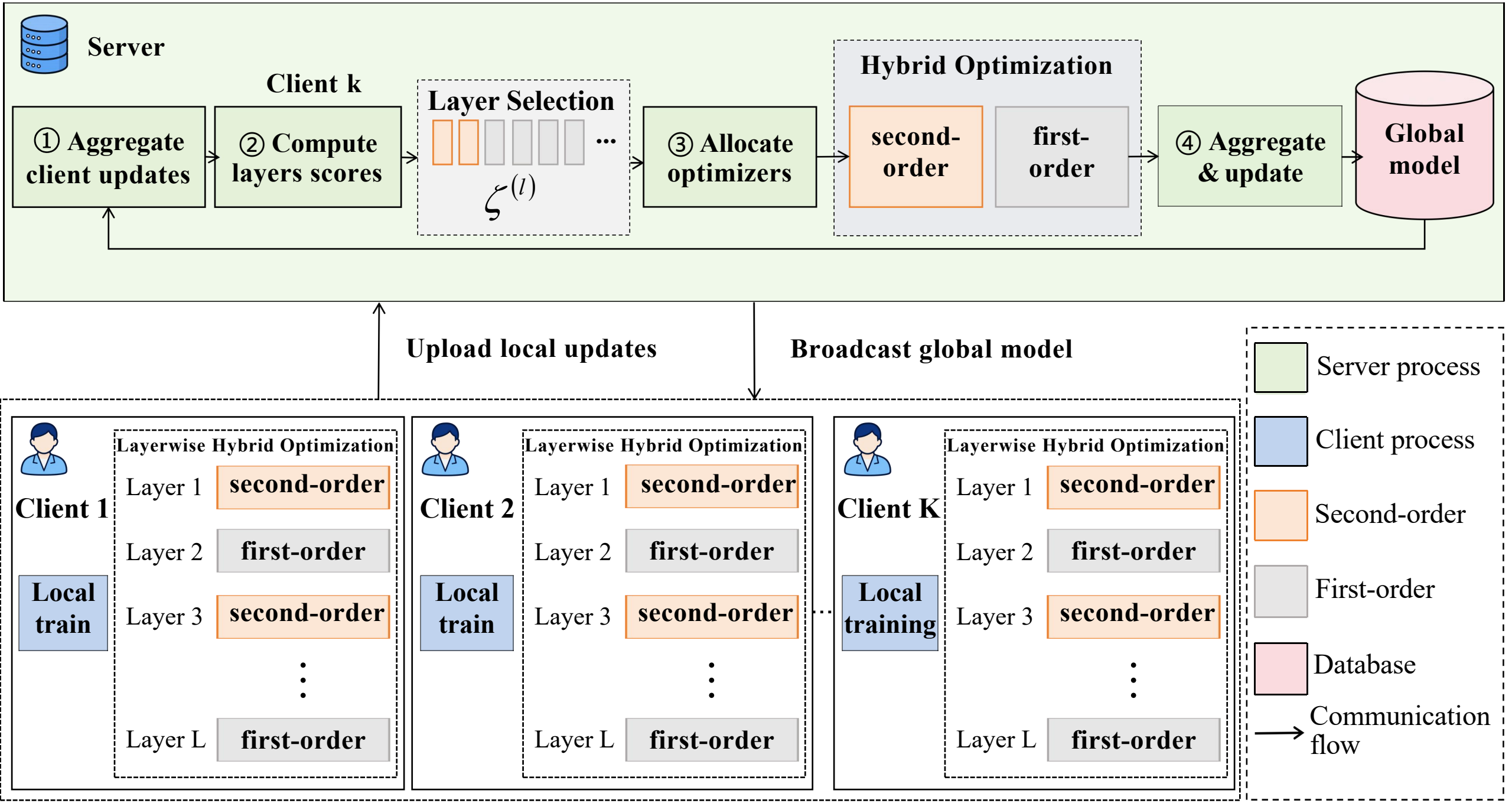

Server
Client k
① Aggregate client updates
② Compute layers scores
Layer Selection
$\zeta^{(l)}$
③ Allocate optimizers
Hybrid Optimization
second-order
first-order
④ Aggregate & update
Global model
Upload local updates
Broadcast global model
Client 1
Client 2
Client K
Local train
Local training
Layerwise Hybrid Optimization
Layer 1
Layer 2
Layer 3
Layer L
second-order
first-order
Server process
Client process
Second-order
First-order
Database
Communication flow